

CIVORA AI

Flood & Healthcare Emergency RAG Knowledge Base

Team HACHOS

Compiled from: WHO | IFRC Red Cross | UNICEF Pakistan | NDMA Pakistan | PDMA Punjab

For RAG-based AI System — Civora AI Healthcare & Emergency Assistant

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CHAPTER 1: Flood Health Risks & Disease Outbreaks

[Source: WHO Flooding & Health Topics | WHO European Region Flood Health Management]

1.1 Overview of Flood Health Impacts

Floods cause both immediate and long-term health consequences. The WHO identifies three categories of health impacts during and after floods:

- **Immediate impacts:** Drowning, traumatic injuries (lacerations, fractures), hypothermia, electrocution from downed power lines, and carbon monoxide poisoning from generators.
- **Short-term impacts (days to weeks):** Waterborne disease outbreaks, vector-borne diseases, respiratory infections, skin diseases, wound infections, and mental health crises.
- **Long-term impacts (weeks to months):** Malnutrition, disrupted healthcare services, damaged infrastructure, displacement-related illnesses, and post-traumatic stress disorder (PTSD).

[Source: WHO, Floods Health Topics, 2022]

1.2 Waterborne Diseases — High Risk During Floods

Floodwater contaminates drinking water sources and spreads fecal pathogens. The following diseases are of highest priority during floods in Pakistan:

Disease	Cause	Symptoms	Risk Level
Cholera	Vibrio cholerae bacteria in contaminated water	Severe watery diarrhea, vomiting, dehydration	CRITICAL
Typhoid Fever	Salmonella typhi via contaminated food/water	High fever, headache, abdominal pain/rash	CRASH
Hepatitis A & E	Fecal-oral route, contaminated water	Jaundice, fatigue, nausea, liver inflammation	CRASH
Acute Diarrhea	Multiple pathogens in floodwater	Loose stools, dehydration, especially in children	CRITICAL
Leptospirosis	Leptospira bacteria in floodwater/surroundings	Fever, muscle pain, headache, jaundice	MODERATE-HIGH
Dysentery	Shigella bacteria or Entamoeba	Bloody diarrhea, fever, cramps	HIGH

[Source: WHO Flood Health Risk Management, European Region]

1.3 Vector-Borne Diseases

Stagnant floodwater creates ideal breeding grounds for disease-carrying mosquitoes. Key threats:

- **Malaria:** Anopheles mosquito bites. Symptoms: high fever with chills, sweating cycles every 2-3 days. Treat with antimalarials. Pakistan has high malaria burden in Sindh and Balochistan.
- **Dengue Fever:** Aedes aegypti mosquito. Symptoms: high fever, severe headache, joint/muscle pain, rash, bleeding gums. No specific treatment — supportive care. Monitor platelet count.
- **Chikungunya:** Same mosquito as dengue. Severe joint pain, fever, rash. Can persist for months.

■ After floods, avoid stagnant water near homes. Use mosquito nets, repellents (DEET-based), and wear long sleeves especially at dawn and dusk.

[Source: WHO Vector Control Guidelines | NDMA Pakistan Flood Health Advisory]

1.4 Skin & Wound Infections

Exposure to floodwater — which contains sewage, chemicals, and pathogens — causes:

- **Wound infections:** Any open wound exposed to floodwater must be cleaned immediately with clean water and antiseptic. Cover with waterproof dressing.
- **Tinea (fungal infections):** Prolonged wet feet cause athlete's foot and other fungal infections. Keep feet dry, use antifungal powder.
- **Cellulitis:** Bacterial skin infection from floodwater exposure. Signs: red, swollen, warm skin. Requires antibiotic treatment.
- **Conjunctivitis (eye infection):** From contaminated water splashing in eyes. Wash eyes with clean water. Avoid rubbing.

■ NEVER drink floodwater. Even small amounts can cause severe illness. Boil all water or use water purification tablets.

1.5 Mental Health Impacts

WHO and UNICEF both recognize mental health as a critical component of flood response. Common conditions:

- **Acute Stress Reaction:** Immediate response to traumatic flood events — shock, confusion, emotional numbness. Usually resolves within days.
- **Post-Traumatic Stress Disorder (PTSD):** Re-experiencing trauma, nightmares, avoidance of flood reminders, hypervigilance. Requires professional support.
- **Grief and Loss:** Loss of family members, homes, and livelihoods. Community support is essential.
- **Children:** May show behavioral changes, bed-wetting, clinging, nightmares. Maintain routines as much as possible.

■ Community listening sessions, peer support groups, and access to safe spaces significantly reduce mental health burden during floods. UNICEF Pakistan deployed mental health support in 2022 floods.

[Source: UNICEF Pakistan HAC Appeal 2022-2023 | WHO Mental Health in Emergencies]

CHAPTER 2: Flood Emergency Response — Step-by-Step SOPs

[Source: NDMA Pakistan | IFRC Disaster Risk Management | WHO Emergency Preparedness]

2.1 Flood Warning Levels & Corresponding Actions

Pakistan's NDMA uses a warning system with different alert levels. Civora AI should respond based on these levels:

Alert Level	Situation	Immediate Actions Required
GREEN (Watch)	Rainfall forecast, possible flooding within 72 hours	Prepare emergency kit. Identify evacuation routes. Check on
YELLOW (Advisory)	Flooding possible within 24-48 hours. Low-lying areas at risk	Move belongings to higher floors. Fill emergency water containers. Charge ph
ORANGE (Warning)	Flooding imminent within 12-24 hours	Evacuate low-lying areas immediately. Move livestock. Turn off electricity/ga
RED (Emergency)	Active flooding occurring. Lives at risk	Call 1122 (Rescue). Do NOT walk in floodwater. Stay on rooftops if trapped.

2.2 BEFORE Flood — Preparedness SOP

Household Level Actions (NDMA Recommended)

- Prepare an **Emergency Kit** containing: drinking water (at least 3 liters per person per day), non-perishable food for 72 hours, first aid kit, flashlight with extra batteries, whistle, dust masks, local maps, and copies of important documents in waterproof bag.
- Identify the **nearest evacuation shelter** in your district. In Punjab: contact PDMA Punjab helpline 0800-02345. Know two exit routes from your home.
- **Secure your home:** Move electrical appliances to higher floors. Store chemicals and medicines above flood level. Clear gutters and drains.
- **Livestock protection:** Move animals to elevated areas. Do not leave animals tied in flood-prone areas.
- **Important documents:** Protect CNIC, property papers, medical records in waterproof bags or upload digitally.
- **Vulnerable family members:** Arrange transport for elderly, disabled, pregnant women, and young children first.

[Source: NDMA Pakistan Flood Preparedness Guidelines | PDMA Punjab SOPs]

2.3 DURING Flood — Emergency Response SOP

If Evacuation Order Issued:

- **Step 1:** Leave IMMEDIATELY when authorities issue evacuation order. Do not wait to see how bad flooding gets.
- **Step 2:** Turn off electricity at main breaker, gas at main valve before leaving.

- **Step 3:** Take your emergency kit. Lock your home. Tell a trusted contact your destination.
- **Step 4:** Use designated evacuation routes only. Avoid roads with water — as little as 15 cm of fast-moving water can knock a person down; 30 cm can sweep away a vehicle.
- **Step 5:** Proceed to designated shelter or move to higher ground.

If Trapped in Flooded Building:

- **Step 1:** Move to the HIGHEST floor immediately. Never go to basement.
 - **Step 2:** Signal for rescue: use bright colored cloth, flashlight, or whistle from windows/rooftop.
 - **Step 3:** Call 1122 (Rescue Punjab) or 1021 (NDMA) and give your exact location.
 - **Step 4:** Do NOT attempt to swim through floodwater unless absolutely necessary.
 - **Step 5:** Stay away from electrical outlets, appliances, and standing water inside building.
- NEVER walk in moving floodwater. 15 cm depth can knock an adult off their feet. Hidden hazards (open manholes, sharp debris, live wires) are invisible under floodwater.

[Source: NDMA Pakistan | IFRC Flood Emergency Response | WHO Flood Health Advisory]

2.4 AFTER Flood — Recovery SOP

- **Return only when authorities declare area safe.** Do not enter flooded buildings until structural safety is confirmed.
- **Inspect for hazards:** Check for gas leaks (smell), structural damage, standing water, downed power lines before entering home.
- **Document damage:** Take photos for insurance and government compensation claims. Contact NDMA/PDMA for registration.
- **Water safety:** Do not use tap water until tested safe. Boil water or use purification tablets. Discard all food that contacted floodwater.
- **Clean and disinfect:** Use bleach solution (1 tablespoon per liter of water) to clean all surfaces. Wear rubber gloves, boots, and masks during cleanup.
- **Seek medical attention** if you have open wounds, develop fever, diarrhea, or any illness within 2 weeks of flood exposure.

■ Register with NDMA/PDMA immediately after floods for relief assistance. Bring CNIC for registration. Pakistan's 2022 flood response registered over 33 million affected people.

[Source: NDMA Pakistan | UNICEF 2022 Pakistan Floods Response Plan]

CHAPTER 3: First Aid Guidelines During Flood Emergencies

[Source: IFRC International First Aid & Resuscitation Guidelines 2020 | IFRC First Aid Policy 2022]

The IFRC's 2020 International First Aid Guidelines — compiled by global experts — provide evidence-based protocols for emergencies commonly encountered during floods.

3.1 Drowning — First Aid Response

Drowning is the leading cause of flood-related deaths. Rapid response is critical.

Recognition:

- Person found in or near water, unconscious or barely conscious.
- Gasping, coughing, or not breathing normally.
- Blue lips/fingernails (cyanosis), cold skin.

First Aid Steps (IFRC Protocol):

- **Step 1 — Safety First:** Do NOT enter water unless trained. Use a rope, branch, clothing, or flotation device to pull the person to safety from the bank.
- **Step 2 — Check responsiveness:** Once on land, tap shoulders and shout. If no response, call for help immediately (1122).
- **Step 3 — Open airway:** Tilt head back gently, lift chin. Look, listen, feel for breathing for no more than 10 seconds.
- **Step 4 — If not breathing — begin CPR:** 30 chest compressions (hard and fast, center of chest) followed by 2 rescue breaths. Continue until help arrives or person starts breathing.
- **Step 5 — Recovery position:** If breathing, place on side (recovery position) to prevent choking. Keep warm.
- **Step 6 — All drowning victims must be seen by a doctor** — even if they appear recovered, secondary drowning can occur hours later.

■ Do NOT attempt to drain water from lungs by holding upside down — this wastes time and causes harm. Begin CPR immediately if not breathing.

[Source: IFRC International First Aid Guidelines 2020, Chapter: Drowning]

3.2 Hypothermia — Cold Water Exposure

Flood victims in cold water lose body heat rapidly. Hypothermia occurs when core temperature falls below 35°C.

Symptoms:

- **Mild:** Shivering, confusion, slurred speech, loss of coordination.
- **Severe:** No shivering, very cold skin, slow/irregular breathing, loss of consciousness.

First Aid:

- Move person to warm, dry environment immediately. Remove wet clothing carefully.
- Cover with blankets including head. Use body heat if blankets unavailable.

- Give warm (not hot) sweet drinks if person is conscious and can swallow.
- Do NOT rub or massage limbs — this can cause cardiac arrest.
- Handle gently — rough movement can trigger dangerous heart rhythms.
- Seek emergency medical care urgently.

■ In Pakistan floods, hypothermia is a risk especially at night and during winter floods. Children and elderly are most vulnerable.

[Source: IFRC First Aid Guidelines 2020 | WHO Flood Health Risk Management]

3.3 Traumatic Injuries — Wounds & Fractures

Wound Management in Flood Conditions:

- **Cuts and lacerations:** Clean wound thoroughly with clean (boiled/purified) water. Remove visible debris carefully. Apply antiseptic (iodine or alcohol). Cover with clean bandage. Change dressing daily.
- **Infected wounds** (red, swollen, pus, fever): Require antibiotic treatment. Seek medical attention immediately.
- **Tetanus risk:** All wounds from flood debris carry tetanus risk. Check vaccination status — if uncertain, seek tetanus toxoid injection within 24-48 hours.
- **Puncture wounds from nails/metal:** High tetanus risk. Do NOT close these wounds — let them drain. Clean well and seek medical care.

Fracture Management:

- Do NOT attempt to straighten broken bones.
- Immobilize the fracture using splints (straight sticks or boards) with padding.
- Secure above and below the fracture site.
- Check circulation (pulse, sensation, warmth) below injury frequently.
- Elevate injured limb if possible to reduce swelling.

[Source: IFRC First Aid Guidelines 2020 | WHO Emergency Trauma Care]

3.4 Dehydration & Oral Rehydration

Diarrheal diseases during floods cause rapid dehydration, especially dangerous for children under 5.

Signs of Dehydration:

- Dry mouth, no tears when crying, sunken eyes, decreased urination, extreme thirst, dizziness.
- **Severe (emergency):** Very sunken eyes, cannot drink, limp, unconscious — requires IV fluids immediately.

Oral Rehydration Solution (ORS) — Recipe:

■ WHO ORS Formula: 1 liter of safe water + 6 teaspoons of sugar + 1/2 teaspoon of salt. Mix well. Give small sips frequently. If ORS packets available, use as directed.

- Adults: 200-400 ml (1-2 cups) after each loose stool.
- Children under 2: 50-100 ml after each stool.
- Children 2-10: 100-200 ml after each stool.

- Continue breastfeeding infants. Do NOT give sugary drinks, juices, or soda as ORS substitutes.

■ A child who cannot drink, is very lethargic, or has bloody diarrhea requires immediate hospital care. Do NOT delay.

[Source: WHO Diarrheal Disease Guidelines | UNICEF ORS Protocol]

3.5 Snake Bites — Elevated Risk During Floods

Floods displace snakes from their habitats into human areas. Snake bites increase significantly during floods in Pakistan.

- **Do:** Keep person still and calm. Remove tight clothing/jewelry near bite. Keep bitten limb at or below heart level. Transport to hospital IMMEDIATELY.
- **Do NOT:** Cut the wound, suck out venom, apply tourniquet, apply ice, or give alcohol.
- Note the snake's appearance for identification if safely possible — but do NOT attempt to catch it.
- Antivenom is available at district hospitals. Time is critical — most deaths occur within 6 hours.

■ During and after floods, wear rubber boots when walking in flood-affected areas. Use a stick to probe ahead when walking in tall grass or debris.

[Source: WHO Snake Bite Guidelines | IFRC First Aid 2020]

CHAPTER 4: Flood Rescue Strategy & Search Operations

[Source: NDMA Pakistan | IFRC Disaster Response | Rescue 1122 Punjab Protocols]

4.1 Pakistan's Official Rescue Framework

Pakistan has a structured rescue system activated during flood emergencies:

- **NDMA (National Disaster Management Authority):** Federal-level coordination. Activates National Emergency Operations Center (NEOC). Coordinates military, provincial, and international response.
- **PDMA (Provincial Disaster Management Authority):** Province-level coordination. Punjab PDMA, Sindh PDMA, KP PDMA, Balochistan PDMA each manage their regions.
- **Rescue 1122 (Punjab):** 24/7 emergency response. Flood rescue boats, trained divers, swift water rescue teams. Call: 1122.
- **Pakistan Army/Navy/Air Force:** Helicopter rescue, boat operations, search and rescue in inaccessible areas. Army deployed 1,973 troops in 2022 floods.
- **Civil Defence & Fire Brigade:** Urban rescue, building evacuation.

[Source: NDMA Pakistan 2022 Flood Response | Pakistan UN Floods Response Plan 2022]

4.2 Emergency Helplines — Pakistan

Service	Number	Coverage
NDMA Emergency	1021	National — 24/7
Rescue (Punjab)	1122	Punjab — 24/7
Police Emergency	15	National — 24/7
Edhi Foundation	115	National — 24/7
Chhipa Welfare	1020	Sindh — 24/7
PDMA Punjab Helpline	0800-02345	Punjab — Toll Free
Pakistan Red Crescent	021-111-177-177	National
Aman Foundation	0317-4288665	Karachi

4.3 Self-Rescue Guidelines — When Trapped

If official rescue has not arrived and you must act:

On Rooftop/High Floor:

- Signal rescuers: Use bright fabric, mirror reflection, flashlight (3 flashes = emergency signal internationally), or whistle (3 blasts = distress).
- Stay visible. Stay together. Do not risk jumping into floodwater.
- Save phone battery — turn on airplane mode between calls.

- Share your GPS coordinates when calling rescue (WhatsApp location sharing).

If Forced to Move Through Shallow Floodwater:

- Use a sturdy pole/stick to check depth and stability ahead.
- Move during daylight only. Never at night — cannot see hazards.
- Stay away from drain covers, bridges, and fast-moving water.
- Keep children on shoulders or use rope attached to adults.
- Remove heavy clothing and shoes if they become waterlogged.

■ A vehicle submerged in 60 cm of water floats and loses steering control. If your car is filling with water: unbuckle, open window (not door — pressure difference), exit through window, swim to surface.

[Source: NDMA Pakistan | IFRC Emergency Response]

4.4 Community-Based Rescue — Volunteer Actions

IFRC and NDMA guidelines recognize community volunteers as the first line of rescue before official teams arrive:

- **Forming rescue teams:** Groups of 4-6 people. Always use rope safety — one person enters water tied to rope held by others on bank.
- **Boat rescue priority order:** 1. Children 2. Elderly 3. Pregnant women 4. Disabled persons 5. Others. One trip per boat — do not overload.
- **Communication:** Designate one person per neighborhood as communication point. Use megaphones or elevated positions to communicate across water.
- **Triage:** Sort rescued people by urgency — unconscious/not breathing first, injured second, ambulatory last.
- **Shelter:** Establish temporary collection points on highest ground. Schools, mosques, and government buildings are priority shelter locations per NDMA.

[Source: IFRC Disaster Response | NDMA Community Resilience Guidelines | Rescue 1122 Protocols]

CHAPTER 5: Water, Sanitation & Hygiene (WASH) During Floods

[Source: UNICEF Pakistan WASH Guidelines | WHO WASH in Emergencies | UNICEF 2022 Pakistan HAC]

WASH (Water, Sanitation, and Hygiene) is identified by UNICEF and WHO as the highest-priority intervention during floods. Contaminated water is the primary driver of disease outbreaks.

5.1 Safe Water During Floods

Water Sources: Safe vs Unsafe

- **UNSAFE during floods:** Tap water (pipes may be contaminated), wells (flooded/contaminated), river/stream water, rainwater collected from roofs, floodwater.
- **SAFER options:** Sealed bottled water, water from raised storage tanks not reached by flood, water distributed by authorities from tankers.

How to Make Water Safe — Step by Step:

- **Boiling (Most Reliable):** Bring water to rolling boil for at least 1 minute (3 minutes at altitude above 2000m). Cool in covered container. Use within 24 hours.
- **Chlorination Tablets:** Follow package instructions. Typically 1 tablet per liter. Wait 30 minutes before drinking. Effective against bacteria and viruses.
- **Bleach (Sodium Hypochlorite):** If 5% bleach — add 2 drops per liter of clear water or 4 drops per liter of cloudy water. Wait 30 minutes.
- **Filtration + Treatment:** Filter cloudy water through clean cloth first, then treat chemically or boil.

■ UNICEF provided safe drinking water to 92,040 people during 2022 Pakistan floods through water tankers and filtration plants. In absence of official support, boiling remains most accessible option.

[Source: UNICEF Pakistan 2022 Flood Response | WHO Emergency Water Treatment]

5.2 Sanitation During Floods

When toilet facilities are flooded or unavailable:

- Use temporary latrines at least 30 meters from any water source and 15 meters from any dwelling.
- Cover waste with soil or ash after each use to prevent fly breeding.
- Designate separate areas for men and women — essential for dignity and safety.
- UNICEF standard: minimum 1 toilet per 20 people in emergency settings.
- Dispose of children's feces in latrines — not on open ground. Childcare areas need special attention.

■ Open defecation near floodwater directly causes cholera and typhoid outbreaks. Even one latrine shared by many is far safer than open defecation.

5.3 Hygiene Practices — Critical During Floods

- **Handwashing:** The single most effective way to prevent disease spread. Wash hands with soap (or ash if no soap) for 20 seconds: before eating, after toilet use, after touching floodwater, before feeding children.

- **Food safety:** Discard ALL food that came in contact with floodwater. Cook food thoroughly. Do not share eating utensils without washing. Prioritize breastfeeding for infants.
- **Body hygiene:** Clean body with safe water as soon as possible after flood exposure. Remove contaminated clothing. Clean any flood-exposed wounds immediately.
- **Menstrual hygiene:** UNICEF emphasizes dignity kits for women during floods. Seek menstrual hygiene supplies from relief distributions.

[Source: UNICEF Pakistan WASH | WHO WASH in Emergencies 2022]

CHAPTER 6: Health Facility & Medical Response During Floods

[Source: WHO Flood Health Checklists | UNICEF Pakistan Health Response | WHO European Region]

During the 2022 Pakistan floods, 501 health facilities in Sindh and 244 in Balochistan were damaged. WHO and UNICEF deployed mobile health teams to compensate.

6.1 Mobile Health Units — What They Provide

- Primary healthcare consultations — fever, diarrhea, respiratory infections, wound care.
- Maternal and neonatal emergency care — deliveries, pregnancy complications.
- Childhood vaccinations — maintaining EPI (Expanded Programme on Immunization) schedule.
- Nutrition screening — mid-upper arm circumference (MUAC) measurement for children.
- Mental health first aid and psychosocial support.
- Medicine distribution — oral rehydration salts, antibiotics, antimalarials, wound care supplies.

[Source: WHO Pakistan Flood Response | UNICEF Health Response 2022]

6.2 Common Medical Protocols in Flood Response

Cholera Case Management:

- Mild-moderate: Oral Rehydration Solution (ORS) — 75 ml/kg over 4 hours.
- Severe: IV Ringer's Lactate 100 ml/kg over 3 hours (adults). Hospital admission required.
- Antibiotic: Doxycycline 300 mg single dose for adults (reduces disease duration).
- Isolation and hygiene: Cholera spreads rapidly — infected persons need separate toilet facilities.

Malaria Case Management:

- Confirmed (RDT positive) *Plasmodium falciparum*: Artemisinin-based Combination Therapy (ACT).
- Confirmed *P. vivax*: Chloroquine + Primaquine (check G6PD status first).
- Severe malaria: IV artesunate — hospital admission urgent.

Acute Respiratory Infections:

- Increased in flood shelters due to overcrowding and cold exposure.
- Pneumonia in children under 5: fast breathing + chest indrawing = danger signs requiring hospital.
- Ensure shelter ventilation. Separate sick from healthy individuals.

■ During 2022 Pakistan floods, WHO supported 656 severely malnourished children in nutrition stabilization centers. Severe acute malnutrition requires specialized ready-to-use therapeutic food (RUTF).

[Source: WHO Disease Control in Humanitarian Emergencies | UNICEF Pakistan 2022]

CHAPTER 7: Vulnerable Groups — Children, Women & Elderly

[Source: UNICEF Pakistan | WHO Flood Health Management | IFRC Disaster Response]

7.1 Children (Under 5)

Children under 5 are at highest mortality risk during floods. One-third of all 2022 Pakistan flood deaths and injuries involved children.

- **Drowning risk:** Children must never be left near floodwater unsupervised. Use ropes, life jackets, flotation devices.
- **Diarrheal disease:** Children dehydrate faster than adults. Monitor closely. ORS immediately at first sign of diarrhea.
- **Malnutrition:** UNICEF estimated 1.6 million children needed treatment for severe wasting after 2022 floods. MUAC < 11.5 cm = severe acute malnutrition (emergency).
- **Vaccination:** Floods interrupt immunization schedules. Measles and polio outbreaks follow. Seek vaccination catch-up at mobile health units.
- **Psychological care:** Maintain routines as much as possible. Keep children with familiar adults. Allow expression of feelings.

7.2 Pregnant and Breastfeeding Women

- Floods increase risk of preterm birth, miscarriage, and maternal mortality due to stress, lack of nutrition, and disrupted healthcare.
- Pregnant women should evacuate first. Avoid floodwater exposure — chemicals and pathogens especially harmful.
- **Danger signs requiring immediate hospital care:** Vaginal bleeding, severe headache, blurred vision, severe abdominal pain, baby not moving, water breaking prematurely.
- Continue breastfeeding during floods — breast milk is safest food for infants and provides immunity.
- Iron and folic acid supplementation must continue — seek from relief distributions or health teams.

7.3 Elderly Persons

- Elderly are at high risk for drowning (mobility limitations), hypothermia (poorer thermoregulation), and medication disruption.
- Priority in evacuation. Assist with mobility. Ensure medications for chronic diseases (diabetes, hypertension, heart disease) are included in emergency kit.
- Dehydration occurs faster in elderly — monitor fluid intake carefully.
- Social isolation in flood shelters increases depression risk. Regular contact with family and health workers essential.

7.4 Persons with Disabilities

- Evacuation must account for mobility limitations. Community must plan specific assistance — designated helpers for each disabled person.

- Hearing/vision impaired persons may not receive evacuation warnings — ensure alerts reach them through personal contact.
- Damaged assistive devices (wheelchairs, hearing aids) need replacement — report to NDMA/PDMA for assistance.

[Source: UNICEF Pakistan 2022 HAC | WHO Flood Health Checklist | IFRC Disability Inclusion]

CHAPTER 8: Pakistan-Specific Flood Context & NDMA Guidelines

[Source: NDMA Pakistan | Pakistan UN Floods Response Plan 2022 | PDMA Punjab]

8.1 Pakistan's Flood Geography — High Risk Zones

Pakistan is among the world's most flood-vulnerable countries due to monsoon rains, glacial melt, and river systems.

- **Highest risk provinces:** Sindh (2022: 89% of houses damaged/destroyed in Pakistan), Balochistan, Khyber Pakhtunkhwa, Southern Punjab.
- **Primary flood rivers:** Indus, Chenab, Jhelum, Ravi, Sutlej. Floods propagate downstream — upstream warnings give communities time to prepare.
- **Flash flood zones:** KP mountain valleys (Swat, Chitral, Dir), northern Punjab, Balochistan hill torrents. Warning time is very short (minutes to hours).
- **Urban flooding:** Karachi, Lahore, Peshawar — poor drainage systems cause rapid urban inundation during heavy rains.

8.2 2022 Pakistan Floods — Scale of Crisis

The 2022 floods were described as Pakistan's worst climate-induced disaster:

Metric	Data
People affected	33 million
People killed	1,760+
People injured	12,867+
Houses destroyed	897,014
Houses damaged	1,391,467
Health facilities damaged (Sindh)	501
Health facilities damaged (Balochistan)	244
Schools damaged/destroyed	22,000+
Children in need of assistance	3.4 million
Economic damage	US\$40 billion
Provinces most affected	Sindh, Balochistan, KP, Punjab

[Source: NDMA Pakistan | UN Pakistan Floods Response Plan | UNICEF HAC 2022-2023]

8.3 NDMA Organizational Structure

- **National level:** NDMA (Chairman appointed by PM) — policy, coordination, international liaison.

- **Provincial level:** PDMA Punjab, PDMA Sindh, PDMA KP, PDMA Balochistan — operational response.
- **District level:** DDMA (District Disaster Management Authority) — ground-level response, shelter management, relief distribution.
- **Relief distribution:** NDMA provides food packs, tents, tarpaulins, mosquito nets, hygiene kits through PDMAAs.
- **International coordination:** NDMA coordinates with UN OCHA, UNICEF, WHO, IFRC for international support.

8.4 Pakistan Climate Vulnerability Context

Pakistan contributes less than 1% of global greenhouse gas emissions but faces disproportionate climate impacts:

- Pakistan is ranked 8th globally in countries most affected by climate risks (UNICEF 2022).
- Pakistan is the 14th most water-stressed country worldwide.
- Glacial lake outburst floods (GLOFs) in northern Pakistan — 3,044 glaciers, 33 potentially dangerous glacial lakes.
- Monsoon rainfall has been 2-3 times the 30-year average in recent years, directly linked to climate change.

[Source: UNICEF Pakistan | NDMA | UN Climate Risk Assessment]

CHAPTER 9: Post-Flood Recovery & Rehabilitation

[Source: NDMA Pakistan | UNICEF Pakistan | WHO Recovery Guidelines | IFRC DRM Policy]

9.1 Health Surveillance in Post-Flood Period

Disease surveillance must be maintained for minimum 3 months after floods:

- **Weekly reporting:** Health workers report cases of diarrhea, malaria, dengue, skin infections from all facilities and mobile teams.
- **Outbreak thresholds:** Even 1 confirmed cholera case = outbreak requiring immediate response. Cluster of fever cases = investigate for malaria/typhoid.
- **Nutrition surveillance:** Monthly MUAC screening of children under 5. Refer SAM cases to stabilization centers.
- **Water quality testing:** Test water sources monthly until declared safe. Report contaminated sources to health authorities.

9.2 Home Cleaning & Disinfection Protocol

- **Wait for floodwater to recede.** Take photographs before cleaning for damage assessment.
- **Protective equipment:** Wear rubber gloves, waterproof boots, N95 mask, eye protection during cleanup.
- **Remove mud and debris** with shovels. Discard all porous materials (mattresses, carpets, insulation) touched by floodwater.
- **Disinfect surfaces** with bleach solution: 1 cup (240ml) bleach per 4 liters of water. Apply, wait 10 minutes, wipe clean.
- **Dry thoroughly** to prevent mold growth. Use fans, open windows. Mold can grow within 24-48 hours in damp conditions.
- **Electrical systems:** Do not restore electricity until a qualified electrician certifies safety.
- **Drinking water:** Flush and disinfect pipes before using tap water. Boil water for drinking for minimum 2 weeks.

9.3 Psychosocial Recovery

IFRC and UNICEF both emphasize long-term psychosocial support as essential for community recovery:

- Establish community support groups — shared experiences reduce isolation and stigma.
- Schools serve as recovery anchors — restoring education quickly supports children's mental health.
- Community events and collective rebuilding efforts restore sense of agency and hope.
- Refer individuals with persistent PTSD, depression, or suicidal ideation to mental health professionals.
- Religious and community leaders play vital role in community healing — engage them in recovery planning.

9.4 Livelihood Recovery

- Agricultural support: Seed distribution, livestock restocking programs through FAO and provincial agriculture departments.
- Cash transfer programs: NDMA/BISP (Benazir Income Support Programme) conditional cash transfers for flood-affected households.
- Registration for compensation: CNIC-based registration at NDMA/PDMA camps. Compensation varies by province.
- Housing: NDMA provides temporary shelter — tents, tarpaulins. Permanent housing reconstruction through Naya Pakistan Housing Authority in flood areas.

[Source: UNICEF Pakistan HAC | NDMA | IFRC DRM Policy | FAO Pakistan]

CHAPTER 10: Emergency Contact Protocols & Quick Reference

[Source: NDMA Pakistan | PDMA Punjab | Pakistan Red Crescent | Rescue 1122]

10.1 Complete Emergency Contacts — Pakistan

Organization	Number	Services	Availability
NDMA Helpline	1021	National disaster coordination, rescue, relief	24/7
Rescue Punjab (1122)	1122	Fire, rescue, medical emergencies, flood rescue	24/7
Police Emergency	15	Law enforcement, security, rescue	24/7
Edhi Foundation	115	Ambulance, rescue, body collection	24/7
Chhipa Welfare (Sindh)	1020	Ambulance, rescue, shelter	24/7
PDMA Punjab Helpline	0800-02345	Flood relief, shelter, evacuation — Punjab	Toll Free
Pakistan Red Crescent	021-111-177-177	First aid, blood, emergency response	24/7
WHO Pakistan	051-2820750	Health emergency coordination	Business hours
UNICEF Pakistan	051-2087300	Child emergency, nutrition, WASH	Business hours
Rescue KP	1122	Emergency rescue — Khyber Pakhtunkhwa	24/7
Sindh Emergency	1020	Sindh provincial emergency	24/7
Aman Ambulance (Karachi)	115	Karachi emergency ambulance	24/7

10.2 Flood Emergency Quick-Decision Guide

This guide helps Civora AI provide instant, appropriate responses:

User Says / Symptom	Civora AI Response Priority	Key Action
'Floodwater is rising'	CRITICAL	Evacuate immediately / Move to highest point
'Someone is drowning'	CRITICAL	Call 1122 / CPR steps if out of water
'Child has severe diarrhea'	URGENT	Start ORS / Go to hospital if severe signs
'I have high fever after flood'	URGENT	Seek medical care / Possible malaria/typhoid
'Snake bit me'	CRITICAL	Go to hospital NOW / Do not cut/suck wound
'Wound from flood debris'	HIGH	Clean wound / Tetanus injection needed
'No safe water'	HIGH	Boil water / Use purification tablets
'Nearest hospital?'	INFO	Use location services to show nearest facility
'Feeling anxious/scared'	SUPPORT	Validate feelings / Provide coping resources
'How to clean my house?'	INFO	Bleach solution / Wait for water to recede

10.3 RAG System Usage Notes

■ This document is optimized for Retrieval-Augmented Generation (RAG). Each section is self-contained and can be retrieved independently. Key terms are emphasized for semantic search. Sources are cited for credibility scoring.

- **Chunk strategy:** Split by section headers (each ## section = 1 chunk, ~200-500 tokens).
- **Metadata tagging:** Tag each chunk with: topic (health/rescue/WASH/mental), urgency (critical/urgent/info), source (WHO/IFRC/NDMA/UNICEF), and language-ready for Urdu translation.
- **Query routing:** Medical symptoms → Chapters 1,3,6. Emergency SOPs → Chapter 2. Rescue → Chapter 4. Water safety → Chapter 5. Pakistan context → Chapter 8.
- **Response disclaimers:** Civora AI should always recommend consulting healthcare professionals for medical decisions. This document provides guidance, not diagnosis.